In this type, the stub axle is attached to the end of the axle by means of king pin. It is used in tractors. As shown in figure 8.12.1 (c).

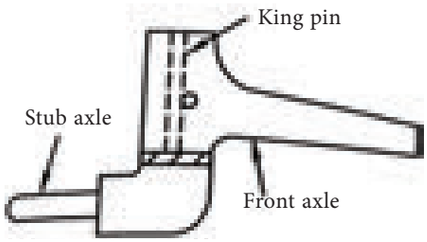


Fig 8.12.1 (c) Lemoine type

4. Reversed lemoine axle

Marmon, American passenger car manufacturer, reversed the arrangement and developed the design. A natural drop to the axle center, is an advantageous feature.

In this type, the stub axle is mounted on the top of the axle by means of king pin. It is used in tractors. As shown in figure 8.12.1 (d).

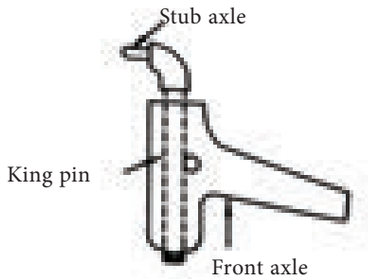


Fig 8.12.1 (d)

8.13 Troubleshooting, cause and remedy in a steering system

8.13.1 Excessive play in steering system

Sl. No.	Possible cause	Remedy
1	Loose / worn out steering gear (rack or box) mounting.	Tighten steering gear mounting bolts / replace mounting bushes.
2	Loose / worn out steering linkages.	Adjust or replace linkages.
3	Worn out of ball joints in steering system	Rectify / replace

8.13.2 Wheel Wobble

Sl. No.	Possible cause	Remedy
1	Mismatched tires or uneven tire pressure.	Inflate tyre upto recommended pressure
2	Unbalanced Wheel	Balance the wheel
3	improper camber angle and toe in	Check the wheel alignment and rectify

8.13.3 Hard Steering

Sl. No.	Possible cause	Remedy
1	Low inflation pressure in the tyres	Properly inflate tyre
2	Improper wheel alignment	Check and rectify the wheel alignment
3	Bend in the front axle	Rectify
4	Misalignment in chassis frame	Rectify
5	Broken / damage bearings and bushes in steering gearbox	Replace

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